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Federated Learning-Based Privacy-Preserving Healthcare Data Classification

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Abstract—Machine learning can support breast cancer diagnosis, but training such models usually requires pooling patient records from several institutions into a central repository, which raises privacy, governance, and regulatory concerns. This study implements and empirically evaluates a federated learning workflow for healthcare data classification in which each participating institution trains locally and only model parameters are exchanged. Five simulated hospital clients are created from the Breast Cancer Wisconsin (Diagnostic) dataset using a Dirichlet partition, and a central server combines the client updates with Federated Averaging (FedAvg). The system is implemented in PyTorch with a purpose-written federated orchestration loop. Under a leakage-free evaluation protocol in which splitting precedes all fitting and resampling, the federated model reaches an accuracy of 96.32 ± 0.39% and an AUC-ROC of 0.9923 ± 0.0029 over five random seeds. For reference, centralized training on the pooled data attains 95.26 ± 0.48% and independently trained per-hospital models attain a mean of 90.49 ± 2.35%. The federated and centralized results are therefore comparable, while both exceed the average isolated-hospital model. We further report convergence behaviour, three client-distribution settings, communication cost, and a gradient-perturbation ablation. We emphasise that the implementation provides no formal differential-privacy guarantee and no secure aggregation, and that exchanged model updates may still leak information about training samples under known attacks.

***Index Terms*—Federated learning, healthcare data classification, privacy preservation, Breast Cancer Wisconsin dataset, federated averaging, non-IID data.**

I. Introduction

Machine learning has become an important tool in clinical decision support, particularly for diagnostic classification tasks such as distinguishing malignant from benign breast masses. Model quality generally improves with the volume and diversity of the training data, which motivates combining records from several hospitals. In the conventional centralized approach, those records are transferred to a common repository before training. This creates a single point of exposure for sensitive patient information and conflicts with the data-governance rules under which many healthcare institutions operate, so hospitals are often unable or unwilling to contribute data to a shared pool.

Federated learning offers an alternative in which the training data never move. Each participating institution trains the model on its own records and transmits only the resulting parameters to a central server, which combines them into a global model and returns it for the next round [1]. Recent surveys report growing interest in applying this paradigm to healthcare [2], [3], while also noting that heterogeneous client data, communication cost, and residual privacy risk remain open problems.

This paper reports a focused empirical study of such a workflow rather than a new federated optimisation algorithm. We implement a FedAvg-based classification pipeline over five simulated hospital clients derived from the Breast Cancer Wisconsin (Diagnostic) dataset, and we evaluate it under an experimental protocol designed so that no test sample can influence training or model selection.

A. Contributions

The contributions of this work are the following.

1) A complete and reproducible implementation of a FedAvg healthcare classification workflow in PyTorch, with all configuration parameters and per-round logs released.
2) A leakage-free evaluation protocol in which the stratified train/validation/test split precedes scaler fitting and minority-class oversampling, so the test partition contains no training sample.
3) A comparison against two reference points that the federated setting is normally claimed to improve upon: centralized training on the pooled data, and local-only training in which each hospital learns in isolation.
4) An evaluation across three client-distribution settings—IID, a non-IID partition with a minimum client size, and an unconstrained Dirichlet partition—reported over five seeds.
5) A round-by-round convergence and communication-cost analysis, including a client-count sweep.
6) An explicit examination of what the implementation does and does not provide in privacy terms, including an ablation of the gradient-perturbation mechanism.

B. Organization of the Paper

The remainder of this paper is organized as follows. Section II reviews related work and states the research gap. Section III describes the methodology. Section IV specifies the experimental setup. Section V presents and discusses the results. Section VI concludes and outlines future work.

II. Related Work

A. Federated Learning in Healthcare

Federated averaging was introduced as a way to train a shared model across decentralized clients by iteratively averaging locally computed parameter updates weighted by the number of local samples [1]. Because raw records stay at the client, the paradigm is frequently proposed for clinical settings where data sharing is restricted. A recent systematic review of methodological work on federated learning for healthcare [2] finds rapid growth in the literature, and a broader survey of federated learning methods [3] catalogues the algorithmic variants developed to cope with heterogeneous clients.

B. Healthcare and Electronic Health Record Applications

A scoping review of federated and distributed learning applied to electronic health records and structured medical data [4] reports predictive-modelling results across a range of clinical tasks, and a review focused specifically on healthcare deployments [5] surveys application areas and implementation considerations. Together these works establish that federated training is technically viable on clinical tabular data and can approach the performance of pooled training in several settings.

C. Privacy and Security Concerns

Keeping raw records local is not equivalent to keeping them private. Zhu et al. [6] demonstrated that shared gradients can be inverted to reconstruct training samples, and Geiping et al. [7] strengthened this result and analysed the conditions under which reconstruction succeeds. Two families of defence are commonly proposed. Differentially private training bounds the influence of any single example by clipping per-example gradients and adding calibrated noise under a formal privacy accountant [8]. Secure aggregation instead uses cryptographic masking so that the server observes only the sum of client updates and never an individual update [9]. Both add computational or accuracy cost, and neither is implied by FedAvg on its own.

D. Limitations of Existing Work

Across the reviews above, several limitations recur. Client data in real deployments are non-IID, and reported results are often sensitive to how heterogeneity is simulated, yet many studies report a single partition and a single random seed. Communication cost is frequently acknowledged but rarely measured. Comparisons are often made only against centralized training, omitting the local-only baseline that represents what an institution could achieve without collaborating at all. Privacy is commonly asserted from the architecture rather than demonstrated, despite the gradient-leakage results noted above. Finally, most studies use benchmark datasets partitioned artificially, so the “hospitals” are simulated rather than real institutions.

E. Research Gap and Positioning

The methods used here—FedAvg on a benchmark tabular dataset—are established, and we make no claim of algorithmic novelty. The gap this study addresses is one of evaluation practice rather than of method. Specifically, we provide a single, internally consistent evaluation that combines elements usually reported separately or omitted: both the centralized and the local-only reference points under an identical protocol; three distinct client-distribution settings; multi-seed results with dispersion; per-round convergence logs; measured communication cost as a function of client count; and a direct, empirical characterisation of the privacy mechanism actually present in the code, including an ablation of its effect. We also adopt an evaluation protocol in which resampling and scaler fitting are confined to the training split, which we found materially affects the measured scores. The intended contribution is therefore a carefully specified and reproducible reference point for a FedAvg healthcare classification workflow, together with an honest account of its limits.

III. Methodology

A. Dataset

We use the Breast Cancer Wisconsin (Diagnostic) dataset, obtained through the `load_breast_cancer` function of scikit-learn, which corresponds to the UCI WDBC collection. It contains 569 samples described by 30 real-valued features computed from digitised images of fine-needle aspirates of breast masses. Each sample is labelled malignant or benign; the class counts are 212 malignant and 357 benign. The dataset contains no missing values, so no imputation is applied. This is the 30-feature diagnostic version and should not be confused with the earlier 9-feature original Wisconsin breast cancer data.

B. Preprocessing and Leakage-Free Evaluation Protocol

Preprocessing follows a fixed order chosen so that no information from the held-out data can reach the model:

1) the raw dataset is split, stratified by class, into 398 training, 57 validation, and 114 test samples.
2) a StandardScaler is fitted on the training split only and then applied unchanged to the validation and test splits.
3) minority-class oversampling is applied to the training split only, enlarging it from 398 to 498 samples with a balanced class ratio.
4) the training split is partitioned across the hospital clients.
5) training proceeds, and the checkpoint is selected using validation accuracy.
6) the selected checkpoint is evaluated once on the test split.

Because both the scaler fit and the resampling occur after the split and touch only training data, the test partition consists of 114 unmodified samples that appear nowhere in training or validation. The class ratio of the test split is left at its natural value (42 malignant, 72 benign); only the training split is balanced.

C. Client Partitioning

The training split is divided among five simulated hospital clients using a Dirichlet distribution over class proportions with concentration parameter $\alpha = 0.5$, a standard way of inducing label skew in federated experiments. The resulting partition is markedly heterogeneous in both client size and class balance, as reported in Table IV. We additionally consider two alternative settings: an IID partition, in which the shuffled training data are split into equal parts; and a constrained non-IID partition, in which Dirichlet proportions are redrawn until every client holds at least 20 samples. The degree of heterogeneity is summarised by the mean ℓ_1 distance between each client's class distribution and the uniform distribution, which is 0.2800 for the main partition.

D. Model Architecture

All clients and the server share an identical feed-forward network. The input layer accepts the 30 standardised features, followed by three hidden blocks of width 128, 64, and 32. Each hidden block applies a linear transform, one-dimensional batch normalisation, a ReLU activation, and dropout with rate 0.3. The output layer is a single linear unit producing one logit; the sigmoid is applied at evaluation time, and training uses binary cross-entropy with logits for numerical stability. The network has 14,785 trainable parameters.

E. Federated Training Procedure

Training proceeds in communication rounds. At the start of round t, the server broadcasts the current global parameters w_t to all participating clients. Each client initialises its local model with those parameters and performs five local epochs of Adam optimisation on its own partition, then returns the updated parameters w_t^k to the server. Raw records are never transmitted. The architecture is shown in Fig. 1 and the round structure in Fig. 2.

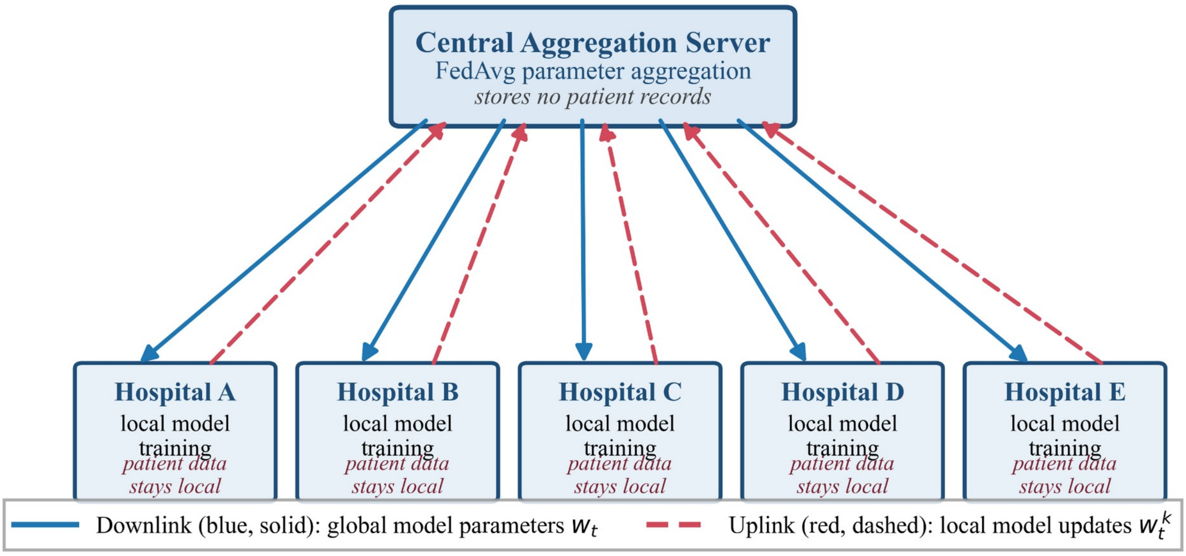

Fig. 1. Federated learning architecture. Five simulated hospital clients train locally and exchange parameters with a central aggregation server that stores no patient records. Blue solid arrows carry the global model parameters w_t from the server to the clients; red dashed arrows carry the locally updated parameters w_t^k back. Aggregation occurs only at the server.

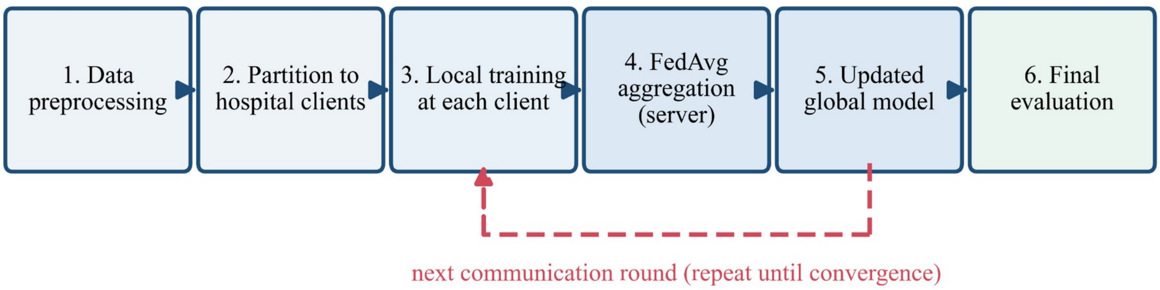

Fig. 2. Workflow of the federated learning pipeline, from preprocessing and client partitioning through local training, FedAvg aggregation at the server, and the updated global model, to final evaluation. The dashed return path indicates the next communication round.

The distinguishing feature of this implementation is its unified evaluation protocol, which combines leakage-free preprocessing, centralized and local-only baselines, multiple client-distribution settings, multi-seed convergence analysis, communication-cost measurement, and an explicit privacy limitation assessment within the same experimental workflow.

F. FedAvg Aggregation

Aggregation is performed by the central server, which is a separate component from every hospital client and holds no patient records. The server computes the sample-weighted mean

$$w_{t+1}=\sum_{k=1}^{K}\frac{n_k}{n}w_t^k,n=\sum_{k=1}^{K}n_k \tag{1}$$

where n_k is the number of training samples held by client k. No hospital aggregates the updates of any other hospital; the hospitals act only as clients. The updated global model is then broadcast for the next round.

G. Privacy-Oriented Gradient Perturbation

The implementation includes an optional gradient-perturbation step applied by each client during local training. After the backward pass and before the optimiser step, the batch-averaged gradient vector is rescaled so that its ℓ_2 norm does not exceed C = 1.0, and Gaussian noise with standard deviation σ = 0.1 is added to each gradient coordinate.

We describe this explicitly as an *experimental Gaussian gradient-perturbation mechanism* and not as differential privacy. It does not satisfy the requirements of a formal (ε, δ)-differentially private algorithm: clipping is applied to the batch-averaged gradient rather than to per-example gradients, so the sensitivity of the aggregate to any single record is not bounded; the noise is not scaled relative to the lot size; and no formal privacy accountant is used. Section V-F reports what this mechanism costs in accuracy.

H. Evaluation Metrics

We report accuracy, precision, recall, F1-score, and AUC-ROC. Precision, recall, and F1 are computed with class-weighted averaging. All metrics are computed on the held-out test split, using the checkpoint selected by validation accuracy. Every configuration is repeated over five random seeds and reported as mean ± standard deviation.

IV. Experimental Setup

The complete configuration is listed in Table I. The dataset, split sizes, partitioning strategy, architecture, and optimisation settings are as described in Section III.

Optimisation uses Adam with a learning rate of 0.001 and weight decay of 1×10^{-5}, a batch size of 32, and five local epochs per communication round. Training runs for at most 50 communication rounds, with early stopping when validation accuracy fails to improve for 10 consecutive rounds; in practice the main configuration terminated between 17 and 27 rounds depending on the seed. Model selection uses validation accuracy only, and the test split is read exactly once per run, after selection.

Each configuration is repeated with seeds 42, 1, 2, 3, and 4. The client-count sweep in Section V-E uses seeds 42, 1, and 2. All experiments were executed on CPU under Windows 11 using Python 3.12, PyTorch 2.11.0, and scikit-learn 1.8.0. The federated orchestration, including the client loop, the server, and the FedAvg aggregation, is implemented directly in PyTorch without an external federated-learning framework.

TABLE I. DATASET AND EXPERIMENTAL CONFIGURATION

Parameter	Value
Dataset	Breast Cancer Wisconsin (Diagnostic)
Source	scikit-learn `load_breast_cancer` (UCI WDBC)
Total samples	569 (212 malignant, 357 benign)
Features	30 real-valued nuclei measurements
Classes	2 (malignant / benign)
Missing values	None
Train / val. / test	398 / 57 / 114 (stratified, split first)
Feature scaling	StandardScaler, fitted on training split only
Class balancing	Oversampling, training split only (398 → 498)
Clients	5 simulated hospitals
Client partition	Dirichlet, α = 0.5
Distribution	Non-IID (non-IID degree 0.2800)
Model	Feed-forward, 30-128-64-32-1
Hidden blocks	Linear → BatchNorm1d → ReLU → Dropout
Dropout rate	0.3
Output layer	Single logit; sigmoid at evaluation
Trainable parameters	14,785
Loss function	Binary cross-entropy with logits
Optimizer	Adam (weight decay 1×10^{-5})
Learning rate	0.001
Batch size	32
Local epochs per round	5
Communication rounds	max 50; early stopping (patience 10)
Aggregation	FedAvg, weighted by client sample count
Model selection	Highest validation accuracy
Random seeds	42, 1, 2, 3, 4
Framework	PyTorch 2.11.0, scikit-learn 1.8.0, Python 3.12
Hardware	CPU (Windows 11, Intel x86-64)

V. Results and Discussion

A. Overall Classification Performance

Table II reports the performance of the federated model in the main configuration. Averaged over five seeds, the global model reaches an accuracy of 96.32 ± 0.39%, precision of 96.37 ± 0.43%, recall of 96.32 ± 0.39%, F1-score of 96.29 ± 0.39%, and AUC-ROC of 0.9923 ± 0.0029. All five metrics are computed from the same validation-selected checkpoint on the same held-out test split, so they describe a single model rather than a combination of different rounds.

TABLE II. CLASSIFICATION PERFORMANCE OF THE FEDERATED MODEL

Metric	Value
Accuracy	96.32 ± 0.39
Precision	96.37 ± 0.43
Recall	96.32 ± 0.39
F1-score	96.29 ± 0.39
AUC-ROC	0.9923 ± 0.0029

B. Comparison with Centralized and Local-Only Training

Table III and Fig. 5 compare three training regimes evaluated under an identical protocol. Centralized training, in which all client partitions are pooled, attains 95.26 ± 0.48% accuracy. Local-only training, in which each hospital trains independently with no aggregation, attains a mean of 90.49 ± 2.35% across the five hospitals. The federated model attains 96.32 ± 0.39%.

The federated configuration achieved 96.32 ± 0.39% accuracy compared with 95.26 ± 0.48% for centralized training. We interpret these as comparable rather than as evidence that federated training is superior. The difference is approximately one percentage point on a test set of 114 samples, so it corresponds to roughly one additional correctly classified sample, and the seed-to-seed intervals of the two regimes overlap. Centralized training also attains the higher AUC-ROC of the two (0.9944 ± 0.0006 against 0.9923 ± 0.0029). The defensible conclusion is that, in this configuration, distributing the training did not degrade classification quality relative to pooling the data.

TABLE III. COMPARISON WITH CENTRALIZED AND LOCAL-ONLY BASELINES

Method	Accuracy	Precision	Recall	F1-score	AUC-ROC
Centralized (pooled data)	95.26 ± 0.48	95.36 ± 0.38	95.26 ± 0.48	95.28 ± 0.46	0.9944 ± 0.0006
Local-only (mean over 5 hospitals)	90.49 ± 2.35	91.43 ± 1.81	90.49 ± 2.35	90.56 ± 2.28	0.9717 ± 0.0095
Federated FedAvg (this work)	96.32 ± 0.39	96.37 ± 0.43	96.32 ± 0.39	96.29 ± 0.39	0.9923 ± 0.0029

The comparison with local-only training requires the same care. The mean across hospitals is 90.49 ± 2.35%, which is clearly below the federated result, and the per-client figures in Table IV show why the mean alone is misleading. Hospital A, which holds 396 of the 498 training samples, reaches 96.49 ± 0.88% on its own—statistically indistinguishable from the federated model. The remaining four hospitals, holding between 5 and 40 samples each, reach between 86.49 ± 6.28% and 92.11 ± 3.77%. In this partition the benefit of collaboration therefore accrues almost entirely to the small clients, while the data-rich client gains little. This also means the strong federated result is substantially influenced by Hospital A: because FedAvg weights each client by its sample count, that single client contributes about 79.5% of the aggregate. This is a limitation of the configuration, and the federated result should not be read as five institutions contributing equally.

TABLE IV. CLIENT PARTITION AND LOCAL-ONLY PERFORMANCE

Client	Samples	Mal./Ben.	Share (%)	Local acc. (%)
Hospital A	396	158 / 238	79.5	96.49 ± 0.88
Hospital B	21	18 / 3	4.2	88.60 ± 4.60
Hospital C	36	33 / 3	7.2	88.77 ± 1.90
Hospital D	5	3 / 2	1.0	86.49 ± 6.28
Hospital E	40	37 / 3	8.0	92.11 ± 3.77

The partition is shown for seed 42; local-only accuracy is averaged over all five seeds. The corresponding distribution of training samples across clients is illustrated in Fig. 6.

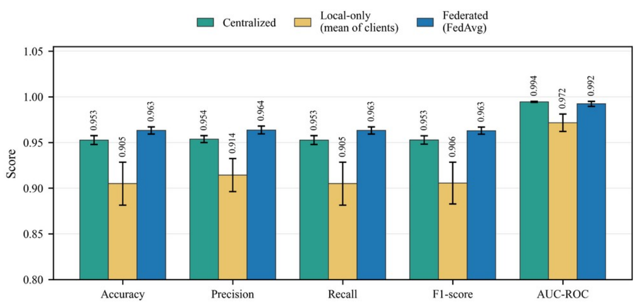

Fig. 5. Classification metrics for centralized, local-only, and federated training. Bars show the mean over five seeds; error bars show ±1 standard deviation.

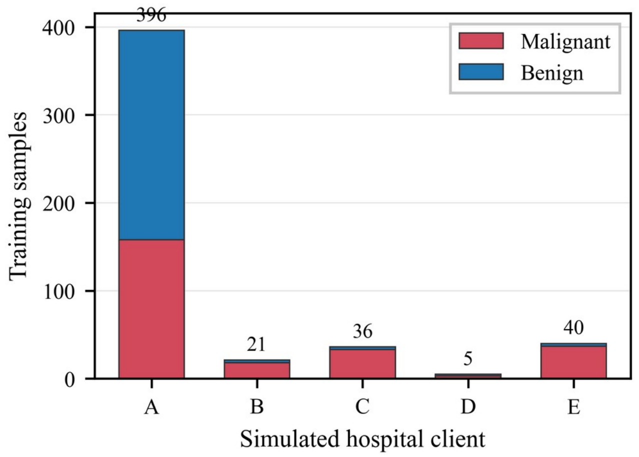

Fig. 6. Training-sample distribution across the five simulated hospital clients for the main non-IID partition (seed 42), separated by class.

C. *Effect of Client Data Distribution*

Table V reports the three partitioning settings. The unconstrained Dirichlet partition reaches 96.32 ± 0.39%, the IID partition 93.51 ± 1.82%, and the constrained non-IID partition with a minimum client size 90.35 ± 2.97%.

These results do not support a simple monotonic relationship in which greater heterogeneity is uniformly better or worse. The ordering is explained by the structure of each partition rather than by heterogeneity as a single quantity. In the unconstrained Dirichlet setting, one client holds most of the data and its internal class balance is comparatively even (158 malignant against 238 benign), so the sample-weighted average is dominated by a client that is itself well conditioned. When a minimum client size is imposed, the sample counts become far more even, but each client's label distribution becomes more skewed, and the averaged model suffers correspondingly. Heterogeneity in client size and heterogeneity in label composition therefore act in opposite directions here, and the aggregate outcome depends on which of the two dominates. We report this as an empirical observation about this benchmark and these partitioning strategies, and we do not generalise it to federated healthcare systems at large.

TABLE V. EFFECT OF CLIENT DATA DISTRIBUTION

Distribution	Accuracy	Precision	Recall	F1-score	AUC-ROC
IID	93.51 ± 1.82	93.68 ± 1.74	93.51 ± 1.82	93.53 ± 1.80	0.9851 ± 0.0071
Non-IID, minimum client size enforced	90.35 ± 2.97	91.34 ± 2.52	90.35 ± 2.97	90.47 ± 2.91	0.9848 ± 0.0083
Non-IID, unconstrained Dirichlet (main)	96.32 ± 0.39	96.37 ± 0.43	96.32 ± 0.39	96.29 ± 0.39	0.9923 ± 0.0029

D. *Convergence Analysis*

Fig. 3 and Fig. 4 show accuracy and loss against communication rounds for the main configuration, averaged over five seeds with a shaded ±1 standard deviation band. Mean test accuracy rises from 0.8737 in round 1 to 0.9491 by round 3 and 0.9596 by round 7, which is within approximately one percentage point of the maximum mean value observed. Beyond round 7 accuracy is essentially flat, moving between 0.9596 and 0.9649 for the remainder of training. The mean global test loss decreases monotonically over the same interval, from 0.5856 in round 1 to 0.1771 by round 10 and 0.1307 at the end of the shortest run, indicating that the model continues to become better calibrated after its accuracy has plateaued.

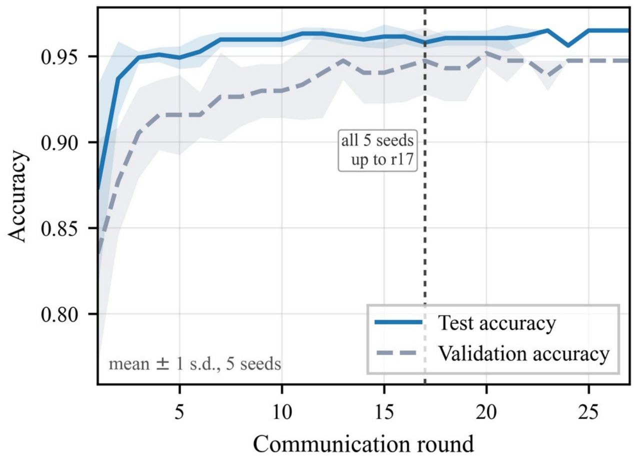

Fig. 3. Global model accuracy against communication rounds, mean over five seeds with a ±1 standard deviation band. The dotted marker indicates the round beyond which fewer than five seeds contribute because of early stopping.

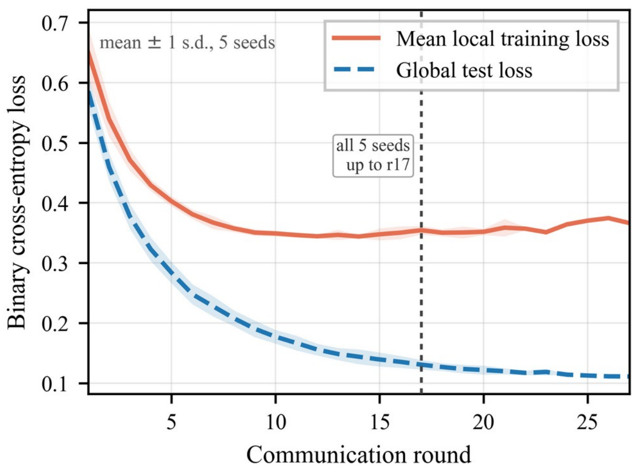

Fig. 4. Mean local training loss and global test loss against communication rounds, mean over five seeds with a ±1 standard deviation band.

Because early stopping terminates each seed independently, the number of contributing runs decreases in later rounds: all five seeds contribute up to round 17, after which between one and four remain, the longest run reaching round 27. The figures mark the round beyond which fewer than five seeds contribute, and the mean and band after that point should be read accordingly.

E. *Scalability and Communication Cost*

Table VI reports a sweep over the number of clients with the total dataset size held fixed, so that a larger client count implies fewer samples per client. Total transmitted volume grows from 2.80 MB with three clients to 23.37 MB with ten, an increase of roughly 8.3×. This growth is the clear and expected finding: each additional client adds a full set of parameter uploads and downloads per round, so communication scales with the client count while the available data does not.

Accuracy across this sweep is not monotonic in the number of clients, and we do not claim that it is. The seven-client setting is markedly weaker than both its neighbours, and this was consistent across all three seeds rather than the effect of a single outlier. Since each client count induces a fresh Dirichlet draw, the partitions differ in geometry as well as in size, and Section V-C already showed that partition geometry has a large effect at this dataset scale. We therefore treat communication cost as the interpretable scalability result and regard the accuracy column as further evidence that, for a dataset of this size, how the data are divided matters more than how many clients they are divided among.

TABLE VI. SCALABILITY IN THE NUMBER OF CLIENTS

Clients	Accuracy (%)	F1 (%)	Comm. (MB)
3	91.23 ± 3.04	91.18	2.80
5	89.18 ± 3.08	89.33	5.18
7	74.56 ± 6.08	74.50	10.10
10	92.40 ± 3.96	92.26	23.37

F. *Privacy and Security Considerations*

Federated learning keeps raw training records at the participating clients: in this implementation no patient record is transmitted, and the server receives only parameter vectors. This addresses the specific risk of pooling records into one repository, but it is not equivalent to a privacy guarantee. Zhu et al. [6] showed that shared gradients can be inverted to reconstruct training samples, and Geiping et al. [7] extended this to more realistic settings. Exchanged model updates may therefore still reveal information about training samples under certain attack settings.

We state plainly what this implementation does and does not provide.

What is provided. Raw patient records remain at the client. Only model parameters are exchanged. An optional Gaussian gradient-perturbation mechanism (batch-level clipping at $C = 1.0$, noise $\sigma = 0.1$) is applied during local training in the reported configuration.

What is not provided. The mechanism above is *not* formal (ε, δ)-differential privacy. Per-example gradient clipping is not implemented, so the contribution of an individual record to the update is not bounded in the sense required by DP-SGD [8]. No formal privacy accountant is implemented, and the quantity the code reports as a cumulative privacy budget has no formal interpretation. Secure aggregation is *not* used: a `SecureAggregator` class exists in the codebase, but it is not invoked in the experimental pipeline, so the server observes each client update individually rather than only their masked sum, in contrast to protocols such as [9]. No defence against malicious clients or a malicious server was evaluated. Accordingly, we make no claim that the system prevents inference from model updates.

Table VII isolates the cost of the perturbation mechanism. Disabling it yields 96.84 ± 1.00% accuracy against 96.32 ± 0.39% with it enabled, so its effect on classification quality is small, roughly half a percentage point, and within the combined seed variation. This should not be read as evidence that the mechanism confers privacy; it quantifies only its accuracy cost.

TABLE VII. EFFECT OF THE GRADIENT-NOISE MECHANISM

Setting	Accuracy	Precision	Recall	F1-score	AUC-ROC
Gradient noise disabled	96.84 ± 1.00	96.88 ± 1.03	96.84 ± 1.00	96.83 ± 1.00	0.9935 ± 0.0023
Gradient noise enabled (as reported)	96.32 ± 0.39	96.37 ± 0.43	96.32 ± 0.39	96.29 ± 0.39	0.9923 ± 0.0029

G. *Limitations*

The following limitations bound the scope of the conclusions.

Dataset. The Breast Cancer Wisconsin (Diagnostic) dataset is a benchmark collection of 569 samples and 30 engineered features. Its size makes the test split small at 114 samples, so differences of about one percentage point correspond to roughly one sample and should not be over-interpreted. It is not a multi-institutional clinical dataset.

Simulated institutions. The five hospitals are produced by partitioning a single dataset. They share feature definitions, measurement protocols, acquisition equipment, and labelling conventions. Real institutions differ in all of these, so the heterogeneity simulated here is label and quantity skew only, and understates the distribution shift of a real federation.

Client imbalance. In the main partition one client holds 79.5% of the training data and another holds five samples. Because FedAvg weights clients by sample count, the global model is strongly influenced by the largest client, and results from this configuration should not be read as a balanced multi-institution collaboration.

Sensitivity to partitioning. Section V-C shows that performance varies substantially with the partitioning scheme, and Section V-E shows a non-monotonic accuracy pattern across client counts. Conclusions drawn from any single partition are therefore fragile at this dataset scale.

Communication overhead. Communication grows with client count and round count, reaching 23.37 MB for ten clients. No gradient compression or quantisation was applied, although the codebase contains configuration entries for such techniques.

Privacy. As detailed in Section V-F, the implementation provides neither formal differential privacy nor secure aggregation, and gradient-leakage risk is not mitigated or measured. No inference or reconstruction attack was mounted against the trained models, so the practical leakage of this configuration remains unquantified.

Robustness. Byzantine or adversarial client behaviour was not evaluated. The codebase contains a Byzantine-detection component, but it was inactive in all reported experiments, and no attack scenario was simulated.

Deployment. All experiments were simulated in a single process on one CPU machine. Network latency, client dropout, asynchronous participation, and the operational and regulatory requirements of real clinical deployment were not modelled.

VI. CONCLUSION

This paper presented an implementation and empirical evaluation of a FedAvg-based workflow for privacy-oriented healthcare data classification. Five simulated hospital clients were derived from the Breast Cancer Wisconsin (Diagnostic) dataset, each training a 30-128-64-32-1 feed-forward network locally in PyTorch, with a central server combining the resulting parameters by sample-weighted averaging. The evaluation protocol places the stratified split before scaler fitting and oversampling, so that the 114-sample test split contains no training data.

Over five random seeds, the federated model reached 96.32 ± 0.39% accuracy and an AUC-ROC of 0.9923 ± 0.0029. Centralized training on the pooled data reached 95.26 ± 0.48% accuracy, while independently trained per-hospital models averaged 90.49 ± 2.35%. The federated and centralized results are comparable; the gap is about one percentage point on a small test set and the seed intervals overlap, so no superiority is claimed. The global model converged quickly, reaching within about one percentage point of its best accuracy by round 7. The comparison across partitions showed that outcomes depend on the structure of the split rather than on heterogeneity alone: the strong result in the main configuration is substantially attributable to one client holding 79.5% of the training data and itself achieving 96.49 ± 0.88% in isolation. Communication volume grew from 2.80 MB with three clients to 23.37 MB with ten clients. These findings motivate future work on larger and genuinely multi-institutional clinical datasets, systematic study of non-IID partitioning and client imbalance, robustness to adversarial clients, and communication-efficient aggregation.

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